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Matrices operation

↳ In order to manipulate and analyze multidimensional data efficiently

Some Matrices operation includes

① Matrix Addition and Subtraction

Add / Sub → corresponding elements of 2 matrices having same dimension

Say I have a Dataset

	Prod A	Prod B	Prod C
Day 1	1	2	3
Day 2	4	5	6
Day 3	7	8	9

} quantity of products that is sold

↓

converting them into matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3\times3} \rightarrow \text{for Store } A \qquad B = \begin{bmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{bmatrix}_{3\times3} \rightarrow \text{for store } B$$

$$A + B = \begin{bmatrix} 1+4 & 2+5 & 3+6 \\ 4+7 & 5+8 & 6+9 \\ 7+1 & 8+2 & 9+3 \end{bmatrix} \rightarrow \begin{bmatrix} 5 & 7 & 9 \\ 11 & 13 & 15 \\ 8 & 10 & 12 \end{bmatrix}$$

② Scalar Multiplication

↳ involves multiplying every element of a matrix by a scalar value

$$\boxed{B = c \times \vec{A}}$$

Eg (Scenario)

↳ Suppose say I have a matrix representing product prices in dollars and we want to adjust these prices for an inflation by a factor of (1.05)

↳ Say my original $P \rightarrow \begin{bmatrix} 10 & 20 & 30 \\ 15 & 25 & 35 \\ 20 & 30 & 40 \end{bmatrix}$

Scalar Multiplication

↓

$$P_{adjusted} \rightarrow 1.05P \rightarrow 1.05 \begin{bmatrix} 10 & 20 & 30 \\ 15 & 25 & 35 \\ 20 & 30 & 40 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 10.5 & 21 & 31.5 \\ 15.75 & 26.25 & 36.75 \\ 21 & 31.5 & 42 \end{bmatrix}$$

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Say I have a dataset (IT Firms)

(in USD)

Base Salary S/w Eng	Base Salary HR	Base Salary Accountant
45K	30K	40K
50K	35K	45K
-	-	-
-	-	-

Suppose say above given prices are for year (2024)

Say I have an inflation of 6% from $\boxed{2024 \rightarrow 2025}$

Multiply the data by a scalar $\boxed{1.06\%}$

③ Matrix Multiplication

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③ ~~Scalar Mat~~ Matrix Multiplication

most of machine learning deep learning algo (Extensively used) matrix mul is used

operation → involves the dot product of rows of the first matrix with columns of the second matrix

Suppose say we have 2 matrices
A → with value (m×n) → no. of columns (11)
B → with value (n×p) No. of rows

The resultant matrix will be $\boxed{C\ (m \times p)}$
rows columns

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

⇒ $(1\times2) + (2\times3) + (3\times4)$
→ $2+6+12$
→ $\boxed{20}$ (Ans)

Suppose say we have 2 matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{(2\times3)} \quad B = \begin{bmatrix} 7 & 9 & 11 \\ 8 & 10 & 12 \end{bmatrix}_{(2\times3)}$$

m×n (n×p)

in order to make this valid to do matrix multiplication

$$B^T = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

So now,

$$B^T \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}_{\substack{(3\times 2) \\ (n\times p)}} \qquad A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{\substack{(2\times 3) \\ (m\times n)}}$$

Now, proper form

$$(m\times n) = (n\times p)$$

$$(2\times 3) = (3\times p)$$

valid for matrix multiplication ↳ my final o/p)

$m\times p \rightarrow \boxed{(2\times 2)}$

Let's see whether, we are able to get it

$$C = A\cdot B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \overset{C_{12}}{(1\times 7)+(2\times 9)+(3\times 11)} & \overset{C_{12}}{(1\times 8)+(2\times 10)+(3\times 12)} \\ \underset{C_{21}}{(4\times 7)+(5\times 9)+(6\times 11)} & \underset{C_{22}}{(4\times 8)+(5\times 10)+(6\times 12)} \end{bmatrix}$$

$$\begin{bmatrix} 7+18+33 & 8+20+36 \\ 28+45+66 & 32+50+72 \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}$$

$\boxed{\text{(final o/p)}}$